Practice Test 3

Nengo and Simple Dynamics

30 minutes · 20 marks · Closed book · No calculator or other devices

Write answers on separate paper. This is a practice simulation, not a prediction of a future test. Use qualitative sketches and symbolic answers. Label axes, components and important levels. No code is required.

1. A vector computation [3 marks]

Five ensembles represent two-dimensional vectors and implement

$$\mathbf{w} = \mathbf{x} - 3\mathbf{y} + 2\mathbf{z} - 2\mathbf{q}.$$

(a) [**1**] Ensemble W has N neurons. Let its filtered activity be the column vector $\mathbf{a}_W \in \mathbb{R}^N$. Write its decoded vector readout using a decoder matrix D, and give the dimensions of D and the readout.

(b) [**2**] Starting with the four represented input vectors and ending at the decoded vector from W, describe the ordered computation performed by the network. Include input encoding, the four connection transforms, their combination, and the final population/readout. About four short lines suffice.

2. Tuning and accuracy [5 marks]

Consider a one-dimensional LIF ensemble. Its normalized intercepts, assigned maximum rates, encoders and random seed are fixed. Nengo recalculates gains and biases when necessary. At radius R, tuning curves are evaluated over the full represented interval $[-R, R]$, and identity decoders are fitted there. The same normalized evaluation points and activity-noise regularization are used in each experiment.

(a) [**2**] On one pair of axes, sketch the firing-rate tuning curve of a positive-encoder neuron with normalized intercept 1/2, first at radius R and then at radius 2R. Label thresholds and rates at the positive domain endpoints. Show the relationship of the two entire curves, not just their endpoints.

(b) [**1**] Give the relationship between radius and absolute representation RMSE under these assumptions, and explain it in one sentence.

(c) [**2**] Instead hold radius fixed and vary either refractory period or membrane time constant, keeping the assigned intercepts and maximum rates fixed as above. State which two tuning-curve features must remain fixed in both experiments. Explain why the interior curve shapes and decoding RMSE can nevertheless change. Do these constraints alone establish a monotonic RMSE trend? About three sentences suffice. Assume all chosen maximum rates are feasible.

3. A feedforward transformation over time [3 marks]

A 100-neuron ensemble A represents a scalar input. A connection from A to a 50-neuron ensemble B computes $y = 1 - 2x$. The connection has a 10 ms low-pass synapse; both decoded probes also use a 10 ms low-pass. The input to A is the pulse $u(t) = 1$ for $1/4 \le t < 3/4$ s and zero otherwise. The network has settled to its baseline before t=0; assume both ensembles represent the required range accurately, with small spiking fluctuations.

Sketch **three aligned traces** from t=0 to 1 s: the input, A's decoded probe, and B's decoded probe. Label the important levels and switching times and show the qualitative transitions. Exact delays, spike times and transient formulas are not required.

4. A neural integrator [7 marks]

The desired state satisfies $\dot{x} = u$, with $x(0) = 0$. The recurrent low-pass time constant is $\tau_r = 50$ ms and the input low-pass time constant is 5 ms. Use the recurrent time constant when selecting connection transforms. The decoded probe uses a 10 ms low-pass. The represented range safely contains the intended state.

(a) [2] For d-dimensional input and state, give A and B in $\dot{\mathbf{x}} = A\mathbf{x} + B\mathbf{u}$, then give the recurrent and input transforms A′ and B′ used by the neural implementation. State their dimensions. No derivation is required.

(b) [3] Take d=1 and use the same input pulse as Q3. Write the definite integral $x(t) = \int_0^t u(s)\,ds$ as a three-case equation. Add one aligned axis beneath your Q3 traces: show the ideal state together with one plausible decoded spiking output through t=1 s. Reuse the input trace already drawn; do not redraw it. Use qualitative deviations, not invented exact spike times.

(c) [2] The experiment is repeated using LIF rate neurons. Explain one fluctuation that should be reduced and two distinct reasons the output can still depart from the ideal integral. About three short sentences suffice; no additional plot is required.

5. Feedback controlled by an input [2 marks]

Here "controlled integrator" means the target system

$$\dot{x} = u + c(t)x,$$

where c is a supplied feedback coefficient with units s^{-1}. Assume matched synapses of time constant τ. Use a state ensemble X and a second ensemble P representing the pair (x, c); idealize the relay into P as instantaneous for this construction.

State the function decoded from P back into X and the input transform into X. Two equations suffice.